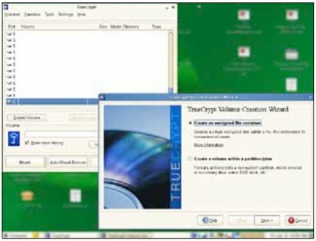

Figure 1: TrueCrypt main screen and volume creation wizard

used to convert encrypted data back to clear text. To keep data safe, one must guard the key.

A hash algorithm is used on data to produce a unique value (a string of unique characters). If the same algorithm is used on modified data, the generated checksum differs, thus confirming the loss of original data integrity. One important property of the hash value is that it cannot be used to recreate the original data.

TrueCrypt uses both these functions: symmetric key encryption and a hash algorithm. Now, let's understand more about TrueCrypt.

Installation

Download the latest stable version available, which is 7.1, for your OS. The basic installation is simple and straight forward; I will walk you through the steps for SuSE 11.1.

Untar the downloaded binaries and launch the set-up. Proceed through various screens and complete the installation by accepting the license. The application browser will show the installed package.

Usage

Launch TrueCrypt, select a slot and click the *Create Volume* button to launch the volume creation wizard. After selecting *Standard TrueCrypt Volume*, enter the desired filename (*DemoVolume*), as shown in Figure 1. Next, select the location where you wish to create the volume. The next screen asks you to select the encryption and hash algorithms. (You may also select a combination of algorithms.) To estimate the encrypted file write speed, you may want to benchmark your system. The test machine had 49 and 51 MBps encryption and decryption speeds. Select the desired encryption and hash algorithm, followed by the volume size, as required. To create *DemoVolume*, I chose AES encryption SHA-512 hash and the size of 10 MB.

Continue and provide a strong pass-phrase (I used *KhulJaSimSimkhuljasimsim*), choose the filesystem type, and format the volume.

Mount this volume (see Figure 2) by selecting *DemoVolume* and entering the corresponding pass-phrase. The mounted volume will be accessible just like any other. Files can be copied to it with a file browser. Once data is copied, simply dismount this volume. Your data is now saved in a TrueCrypt encrypted volume file, which you can email, copy on a pen drive, or even synchronise with cloud-based storage services.

Figure 2: Mount volume

You can use this tool to encrypt complete partitions, a pen drive or any data storage device to secure your data, just like creating a new volume of the required size. Do use this utility, and let your friends know about it. Also, do consider making a donation to this fantastic project. END

References

[1] TrueCrypt website: *www.TrueCrypt.org*
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[3] CNET weblink: *http://forums.cnet.com/7726-6132_102-3330925.html*

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